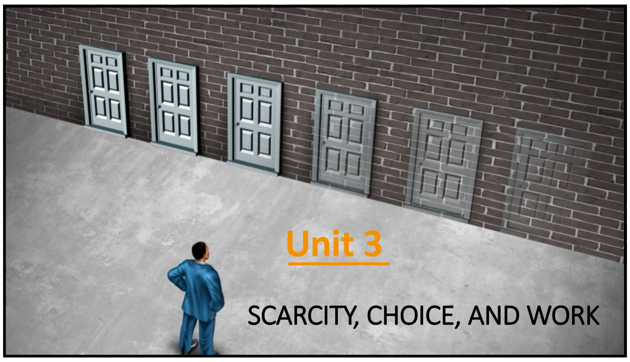


1

Important Concepts

- Preferences
- Utility
- Constraints
- Labour and production
- Trade-off
- Opportunity costs
- Feasible frontier (FF)
- Decision-making and scarcity
- Hours of work and economic growth
- Income and substitution effects

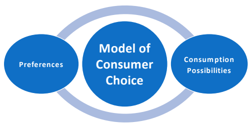
2

The Context for Unit 3

- Model of consumer choice (slides only)
- Labour is an input in the production of goods and services. (Unit 2)
- New technologies raise the productivity of labour. (Unit 3)
  - How would that affect living standards/our income?
  - How would that effect the choices individuals make about the goods to consume?
  - How would that affect the free time and working hours chosen by individuals?

3

ECONOMIC MODEL: DECISION-MAKING UNDER SCARCITY



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Decision-making under scarcity

- Decision-making under scarcity is a common problem because we usually have **limited means** available to meet our **needs**.
- Economists model decision-making under scarcity by:
  - defining all of the feasible actions (i.e. what is possible)
  - evaluating which of these feasible actions is best for the decision-maker (assuming the decision-maker is rational)
- We introduce a model of decision making under scarcity to answer the questions:
  1. What is the optimal basket of goods to choose given our preferences and income, as well as prices?
  2. How much time should we spend working when facing a trade-off between more free time and the potential to earn more income?
- This model also helps to explain differences in the hours that people work in different countries, and the changes in our hours of work throughout history.

This decision-making model will be used repeatedly throughout this course.

5

Model of Individual consumer choice

- Let's first introduce a model of consumer choice to better understand how an individual determines which goods they buy
  - ➢ Then in the next lecture we extend this to look at how much time an individual chooses to spend working, when facing constraints
- We consider:
  - an individual's **preferences**
  - an individual's **consumption possibilities** as constrained by their income and prices

Preferences → CHOICES ← Consumption Possibilities

6

Preferences

- A consumer can compare any two possible combinations of goods A and B so that the consumer
  - prefers A to B
  - prefers B to A
  - is indifferent between A and B
- We assume that a rational decision-maker's preferences are **consistent**

7

Preferences and Utility

Utility

- A numerical indicator of the value that one places on an outcome
- Higher valued outcomes (i.e. those offering greater utility) will be chosen over lower valued ones (i.e. those offering less utility) when both are feasible.
  - If consumer prefers A to B ➔ utility (number) of A must be higher than that of B
    e.g. utility of A = 10 and utility of B = 5
  - If consumer prefers B to A ➔ utility of B must be higher than that of A
  - If consumer is indifferent between A and B ➔ utility of A and B must be the same

8

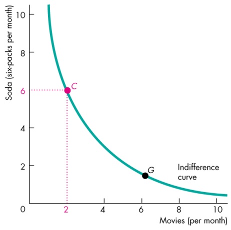
Preferences and Utility(cont.)

- Utility is a measure using an ***arbitrary scale***
- The numerical values have ***no intrinsic meaning.***
- ***Important*** thing is that the utility numbers must reflect the consumer's preferences.
- Generally, more consumption gives more **utility**.

9

Preferences and Indifference Curves

- We can also represent preferences graphically using **indifference curves**.
- An indifference curve shows the points which indicate the **combinations of goods** that provide a **given level of utility** to the individual among which the individual is **indifferent**.
- Figure (a) illustrates all those points which show the combination of soda and movies that provide the same level of utility as point C.
- For example, an individual is indifferent between C and G.

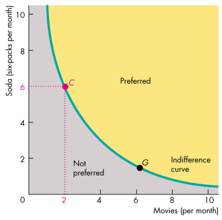


(a) An indifference curve

10

Preferences and Indifference Curves

- All the points above the indifference curve are preferred to the points on the curve.
- All the points on the indifference curve are preferred to the points below the curve.

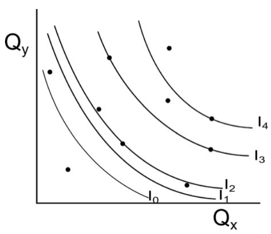


(a) An indifference curve

11

Preferences and indifference curves

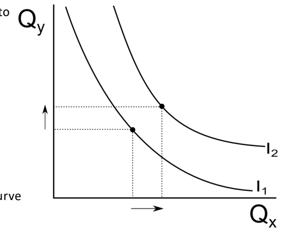
- One curve is not enough to completely represent someone's preferences.
- Any number of indifference curves can be drawn for a person so that **any two** points can in principle be compared.
- Every point lies on **some** indifference curve
- Higher indifference curves are preferred over lower ones.
- Such a system of indifference curves work like contour lines on a map and is called a **preference map.**



12

Properties of indifference curves

- Indifference curves slope downward due to **trade-off** between two choices. (You can compensate a person for the loss of x by giving more y and vice versa.)
- **Higher indifference curves** correspond to **higher levels of utility.**
- Indifference curves are (usually) **smooth**.
- Indifference curves **never cross**.
- As you move right along an indifference curve its **slope gets flatter** – it is **convex**.
- The slope of the indifference curve is the **marginal rate of substitution (MRS)!**



13

The marginal rate of substitution (MRS)

- The (absolute) magnitude of the slope at any point on an indifference curve is the marginal rate of substitution (MRS).
- The **MRS** is the **rate at which a person is willing to give up good y** (measured on the y-axis), to get an **additional unit of good x** (measured on the x-axis) in order to **remain on the same indifference curve**.

 (same indifference curve means same level of utility – consumer remains equally satisfied)

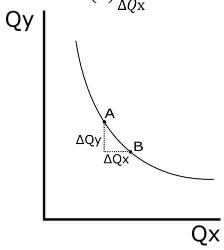
$$\frac{\Delta y}{\Delta x} = \frac{\Delta Q\text{sodas}}{\Delta Q\text{movies}} = (-)MRS$$

14

The marginal rate of substitution (MRS)

- A person that moves from A to B obtains ΔQ_x and loses ΔQ_y **but is just as "well off" as before**
- ΔQ_y is the **most** that this person is **willing** to give up to get ΔQ_x
- $\Delta Q_y/\Delta Q_x$ is the **rate** at which he/she will sacrifice y in order to get more x
- If $\Delta Q_y/\Delta Q_x$ is high (i.e. steep slope & high MRS), it is clear that he/she will prefer product x relative to product y, and will therefore give up more y for a given ΔQx

$$(-)\frac{\Delta Qy}{\Delta Qx} = MRS$$



15

Diminishing marginal rate of substitution

- People generally prefer to consume a balanced bundle of goods.
- People are generally more willing to give up (trade away) goods that they have in abundance and less willing to give up goods that are scarce.
- These differences in MRS cause the indifference curve to be **convex**.
- **Diminishing MRS**
  - general tendency for a person to be willing to give up less of **y** to get one more unit of **x** — whilst remaining equally satisfied — as the quantity of good **x** she has increases.

16

Model of Individual consumer choice

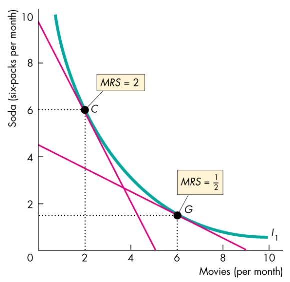
Preferences

Consumption Possibilities

CHOICES

17

Diminishing marginal rate of substitution



For example, as the consumption of movies increases that of soda decreases.
Therefore MRS decreases.

18

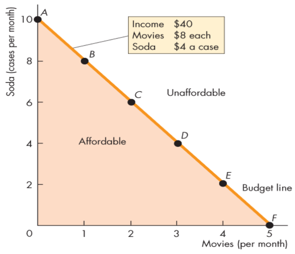
Consumption Possibilities

Previously, we saw how the consumers preferences can be represented by indifference curves.

We now introduce the **constraints** faced by a consumer.

The amount of goods that a consumer can purchase over a given period of time is limited by the consumer's income and by the prices of the goods.

The consumer faces a **budget constraint**, represented by the budget line.



19

Budget line equation

- The budget line represents all combinations of goods that could be consumed if an individual chose to exhaust all of their budgetary resources (income)
- That is, the budget constraint satisfies the condition that the **maximum level of consumption is equal to income**, or:

$$Y = p_x . q_x + p_y . q_y$$

where p_x and p_y are the prices of the two goods, *X* and *Y*, and q_x and q_y are the quantities consumed, and $p_x . q_x + p_y . q_y$ is total expenditure.

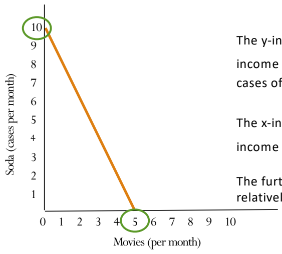
- The budget line is plotted with q_y as the dependent variable, and so rearranging the budget constraint to have q_y on its own on the LFS of the equation gives us:

$$q_y = \frac{Y}{p_y} - \frac{p_x}{p_y} . q_x$$

20

Budget line equation

$$q_y = \frac{Y}{p_y} - \frac{p_x}{p_y} . q_x$$



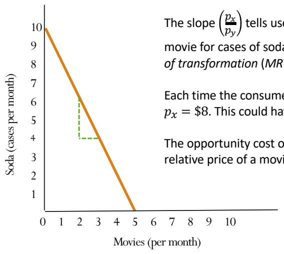
The y-intercept $\left(\frac{Y}{p_y}\right)$ gives us the consumer's real income in terms of cases of soda i.e. the amount of cases of sodas that could be bought with $40.

The x-intercept $\left(\frac{Y}{p_x}\right)$ gives us the consumer's real income in terms of movies.

The further away from the origin the intercept/s, the relatively "wealthier" the consumer is.

21

Budget line equation $q_y = \frac{Y}{p_y} - \frac{p_x}{p_y} q_x$



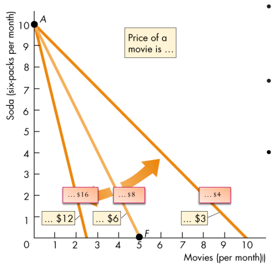
The slope $\left(\frac{p_x}{p_y}\right)$ tells use that rate at which the consumer **must** trade each movie for cases of soda (in Unit 3 this is referred to as the *marginal rate of transformation* (*MRT*))

Each time the consumer purchases a movie ticket, they are spending $p_x = \$8$. This could have been spent on 2 cases of soda ($p_y = \$4$).

The opportunity cost of a movie is 2 cases of soda. This is also the relative price of a movie. What is the relative price of a case of soda?

22

A Change in Prices

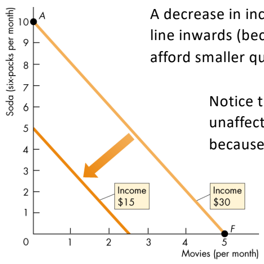


(a) A change in price

- A change in the price of the good changes the slope (i.e. relative price) of the budget line, ceteris paribus.
- The figure shows the rotation of a budget line after a change in the relative price of movies.
- Money income remains $40

23

A change in (real) income



(b) A change in income

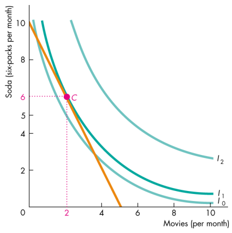
A decrease in income shifts the budget line inwards (because the consumer can afford smaller quantities).

Notice that the slope of the budget line is unaffected by a change in money income because the relative prices stay the same.

24

Consumer choice

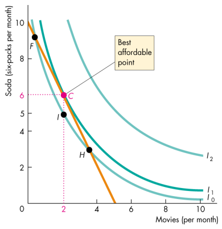
- We now bring together the **preferences** of the consumer (given by their **indifference curves**) and the **income and price constraints** faced by the consumer (given by their **budget line**)
- This allows us to examine how the consumer chooses which goods to purchase and in what quantities to maximise utility
- Here the best affordable point is C.
- Why?



25

Consumer choice

- Points F, I, C and H are all **affordable** (why?)
  - At **point F** Lisa can consume more soda and see fewer movies than at C.
  - At **point H**, she can see more movies and consume less soda than at C.
  - At **point I**, she consumes less soda than but see the same number of movies as at C.
- **She is indifferent between F, I, and H** (why?)
- **She prefers C to I** (why?)
- **She prefers C to either F or H** (why?)



26

Consumer choice

The consumer's best affordable (optimal) point is:

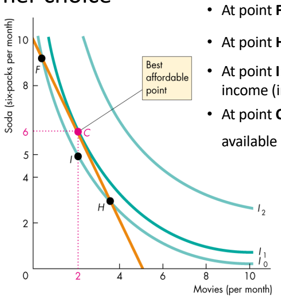
- On the budget line
  - Spends all available money income
- On the highest attainable indifference curve
  - Maximizing utility
- Has a marginal rate of substitution between the two goods equal to the relative price of the two goods (why?)

$$MRS = \frac{P_x}{P_y}$$

27

Consumer choice

3.4

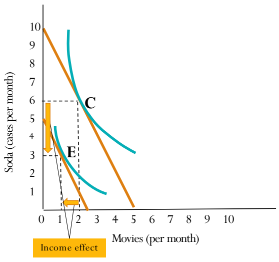


- At point **F**: MRS > $\frac{P_x}{P_y}$
- At point **H**: MRS < $\frac{P_x}{P_y}$
- At point **I**: Lisa does not spend all available income (inside budget line)
- At point **C**: MRS = $\frac{P_x}{P_y}$ and Lisa spends all available income

28

Income effect (due to income change)

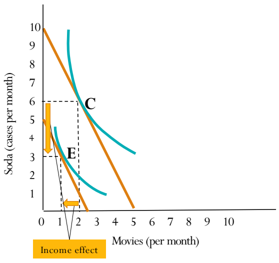
- Graphically, a decrease in income causes the budget line to shift parallel inwards
- Why?
- The consumer's real income in terms of both goods has fallen
- Prices have not changed, which means the slope of the budget line remains unchanged
- The income effect results in a movement from point C to point E
  - To a lower indifference curve
- The income effect is a decrease in movies by 1, and a decrease in cases of soda by 3.



29

Normal vs Inferior goods

- The quantity consumed of a normal good tends to increase as a consumer's income rises.
- The quantity consumed of an inferior good tends a fall as a consumer's income rises.
- Movies and cases of soda are both normal goods.



30

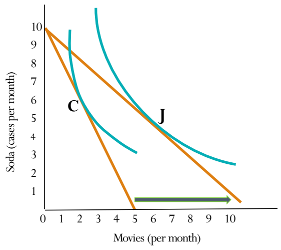
Consumer choice: The price effect

- What would happen if the price of one of the goods changes?
- For example, let's say the price of movies fall but the price of sodas and the income of the consumer remain unchanged.

31

The Price Effect

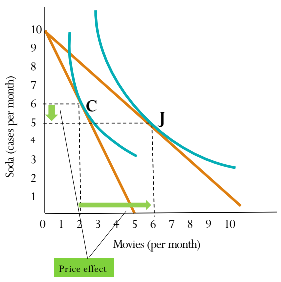
- Graphically, the decrease in the price of movies causes the budget line to swivel outwards along the axis for movies.
- There is no change in the y-axis intercept
- This is because the consumer's real income in terms of movies has increased, but their real income in terms of sodas has remained the same.



32

The Price Effect

- Also, if the slope of the budget line is $\frac{P_x}{P_y}$, and Px has fallen, then the relative price of a movie has fallen and the budget line becomes flatter
- The price effect is the movement from the original optimal consumption point C to the new optimal consumption point J
- The price effect is a decrease in sodas of 1 case, and an increase in movies by 4.



33

Consumer choice: The income and substitution effect

- The **price effect** can be divided into **two effects**:
  1. **An income effect**
     - The effect of an increase in (real) income holding relative price constant
  2. **A substitution effect**
     - The effect due to a change in relative prices

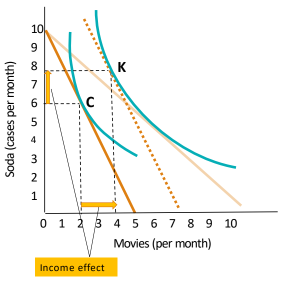
34

- **The Income Effect:** the change in consumption that results when additional (or less) income moves the consumer to a higher (or lower) indifference curve, holding relative prices constant.
- A consumer's purchasing power (i.e. real income) changes when one of the prices change; for example the consumer is able to purchase more of both goods if price decreases.
- Purchasing power could have similarly increased had the consumer received more (nominal) income whilst keeping prices the same
- The income effect therefore represents the effect that a price change has on purchasing power, but treats it as if this hypothetically occurred through a change in income.
- This means we represent it graphically by a parallel shift of the budget line (i.e. we assume relative prices haven't changed)

35

The Income Effect

- The move from C to K is the **income effect.**
- This is found through plotting a hypothetical budget line that:
  - has the same slope as the original budget line, and
  - is tangent to the higher indifference curve
- The income effect is positive for both soda *and* movies



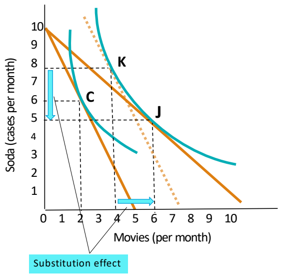
36

- **The Substitution Effect:** the effect of a change in price on the quantity bought given the new level of utility.
- The substitution effect captures the change in consumption that occurs as a result of a price change of the one good that changes the relative price
- For example, if the price of movies falls, movies are now relatively cheaper than before, but soda is relatively more expensive
- A 'rational' consumer would *substitute away* from the relatively more expensive soda *toward* the relatively cheaper moves
- The substitution is indicated by a movement along the indifference curve.

37

The Substitution Effect

- The move from K to J is the **substitution effect.**
- This is found through moving along the higher indifference curve towards the point where the new budget line is tangent to the higher indifference curve
- The substitution effect is positive for quantity of movies, but negative for quantity of soda

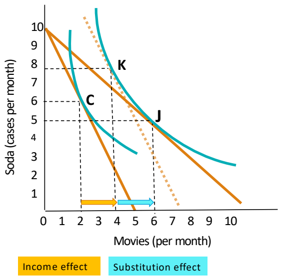


38

The Income and the Substitution Effect

The impact of a price decrease in movies on the consumption of movies:

- **Income effect:** quantity consumed increases (real income increases, and the consumer consumes more of the product because it is a normal good).
- **Substitution effect**: quantity consumed increases (the consumer consumes more of the product that is relatively cheaper).

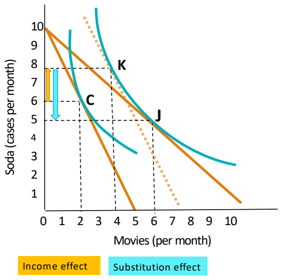


39

The Income and the Substitution Effect

The impact of a price decrease in movies on the consumption of soda:

- **Income effect:** quantity consumed increases (real income increases, and the consumer consumes more of the product because it is a normal good).
- **Substitution effect**: quantity consumed decreases (the consumer consumes less of the product that is relatively more expensive).



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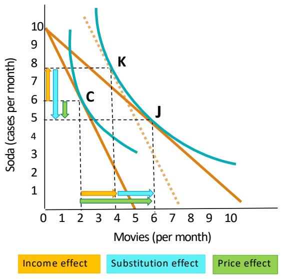
The Income + Substitution Effect = Price effect

The impact (price effect) of a price decrease in movies on the consumption of movies is positive.

The impact (price effect) of a price decrease in movies on the consumption of sodas is negative.



41

Back to the CORE 2.0

42

3.1

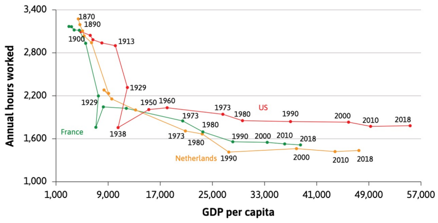
How many hours will you work?

- Assume you work for $15/hour for 40-hour working week
  - Earn $600 per week
- Now get the opportunity to work for $90/hour
  - Still work 40 hours and earn $3 600 per week?
  - Work only 7 hours and earn $630 (about the same as before)?
- What happened in reality over time?

43

3.1

Figure 3.1 Annual hours of work decreased



44

3.2

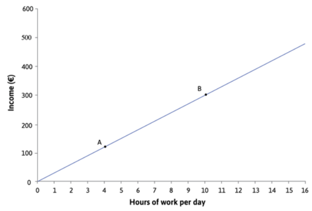
Scarcity

- Economics is the science of making choices between alternatives, because we don't have the means to satisfy all our wants.
- How much time will you work?
  - It is a function of wages
  - It will also depend on your preferences
  - Must make a choice of income vs free time
- Your income y = wages (w) x Hours worked (h)

45

Figure 3.2 Income vs hours worked

3.2

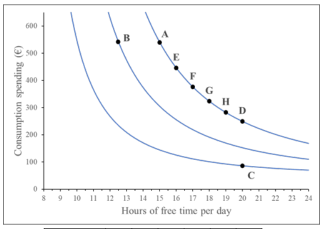


Karim earns €30 per hour. Maximum of 16 hours, because he has to sleep, eat, etc.
If he works 4 hours, he will earn €120 (A)
Slope = $\Delta Y/\Delta L$ = 120/4 = 30 (equal to the wage)
Karim's scarcity problem: Wants both high levels of income and free time, which is impossible

46

Figure 3.4 Karim's choices between consumption and free time

3.3

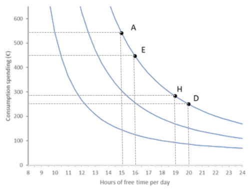


	A	E	F	G	H	D
Hours of free time	15	16	17	18	19	20
Consumption (€)	540	446	376	323	282	250

47

Figure 3.5 Marginal rate of substitution

3.3



48

Working hours and free time

3.8

- The two-good consumption model can also be used to analyse a worker's choice on the number of hours to work (labour supply)
- Suppose you earn a wage of w for every hour that you work.
- This is the amount of *consumption* you can enjoy per hour of work.
- But it is also the opportunity cost of an hour of *free time,* i.e. an hour spend doing something else.
- The two "goods" being consumed in this model are
  - 1) *consumption goods* (bought with the wage money earned) and
  - 2) *free time*.
- You have a "budget" of 24 hours per day.

49

Working hours and free time

Recall from earlier in Unit 3: **Budget constraints** are the feasible frontiers (FF) for *consumption choices.*

The equation of the budget constraint in the model of consumer choice was:

$$q_y = \frac{Y}{p_y} - \frac{p_x}{p_y}.q_x$$

The equation of the budget constraint in a model of labour supply (i.e. how many hours to work) is:

$$\mathrm{c} = w(24 - t)$$

c = maximum level of consumption
w = wage
t = free time (24 – t = hours worked)

50

Working hours and free time

$$\mathrm{c} = w(24 - t)$$

c = maximum level of consumption
w = wage
t = free time (24 – t = hours worked)

"The budget constraint represents all combinations of goods and services that one could acquire that exactly exhaust one's budgetary resources."

That is, it describes using all money earned through labour, $w(24 - t)$, for consumption, c.

51

3.4

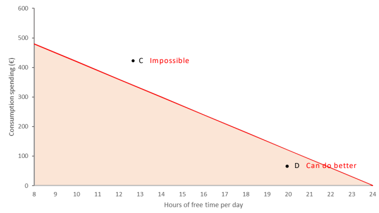
Feasible set

- Karim wants his consumption and free time to be as high as possible
- More free time means less consumption
- If he works *h* hours
- Income *y* = *wh*
- If he takes *t* hours free time
  - *y* = *w*(24 – *t*)
- Figure 3.6 assumes he earns €30/hour
  - *y* and consumption *c* = 30(24 – *t*)
  - Cannot work more than 16 hours – has to sleep, eat. etc.

52

3.4

Figure 3.6 Feasible set and budget constraint



Hours of work	0	2	4	6	8	10	12	14	16
Free time, *t*	24	22	20	18	16	14	12	10	8
Consumption, *c* (€)	0	60	120	180	240	300	360	400	480

53

3.4

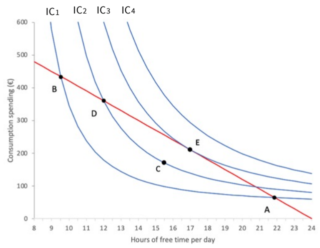
Feasible set

- Slope of the budget line = $\frac{\textit{Change in consumption}}{\textit{Change in hours worked}}$
- $= -\frac{480}{16}$
- = - 30 (which is the hourly wage rate)
- How much consumption must be given up for an extra hour of free time
- Slope of the budget line is the marginal rate of transformation (MRT)
- **MRT**: Trade-off Karim is willing to make between consumption and free time. The trade-off Karim is constrained to make by the feasible frontier

54

3.5

Consumer equilibrium



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3.6

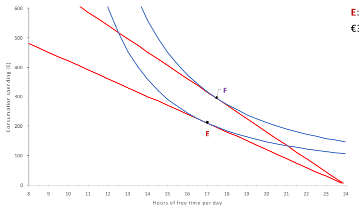
Technological progress and hours worked

- New technology raises productivity of labour
- If labour has bargaining power, it will raise their wages
- How does increasing wages affect living standards and hours worked?

56

3.6

Fig 3.9 Technological progress and hours worked



E: Karim starts with a wage of €30/hour and works 7 hours

Assume his wage increases to €45/hour his feasible set expands

Karim's new budget line will intersect the consumption axis at 720 hours and free time axis at 24

F: Will be the point Karim chooses `now as it gives him more utility than at point E

Karim's MRS between free time and consumption is equal to the MRT between them, that is also equal to the wage (now higher at 45)

57

3.6

Technological progress and hours worked

- Karim's budget constraint moves outwards, expanding his feasible set. He can now afford combinations of consumption and free time that he could not afford before.
- This is the **income effect**: He can now consume more of the goods he values
- The increase in wages increases the opportunity costs of free time and gives him a greater incentive to work more.
- This is the **substitution effect**: He now wants to work more to increase his consumption, rather than having more free time.
- In Figure 3.9 the net effect of the wage increase is that Karim works fewer hours. The income effect is thus larger than the substitution effect

58

3.7

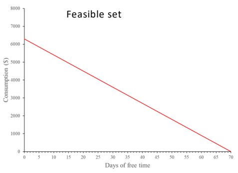
Income and substitution effects

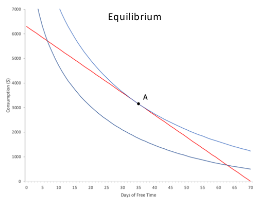
- You take on a part-time job after your exams at the end of the year until classes start again next year
- It will be for 70 days and you care about the consumption that you can enjoy from your earnings and the number of days of free time
- If w is your daily wage and d the number of free days:
  - Consumption $c = w(70 - d)$
- You start with a daily wage of $90 per day

59

3.7

Income and substitution effects





Initially equilibrium at **A**, where you choose 34 free days during the break and earn $3 240 ($90 x 36) for the 36 days of work. There MRT = MRS = w

60

3.7

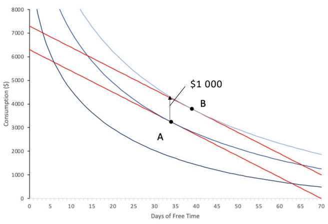
Income and substitution effects

- Due to passing Economics 114 with distinction, you receive a cash payment of $1 000 from the university
- You can spend this money as you like
- This means your budget constraint shifted upwards by $1 000
- Your budget constraint is now:
  - $c = 90(70 – d) + 1\,000$
- Each hour of free time still reduces your consumption by $90($w$)
- What will happen to your consumption?

61

3.7

Income and substitution effects



Increase of income by $1 000 shifts budget constraint up by $1 000

The slope of the budget constraint stays the same as all prices stays the same

Equilibrium shifts to point **B**, where once again MRT = MRS = w

You choose not to work for 39 days (work 3 days less), earn $2 790 and spend $3 790

This is the **income effect**: the increase in consumption if earnings increases. Note that consumption does not increase by $1 000, because you choose to work fewer days.

62

3.7

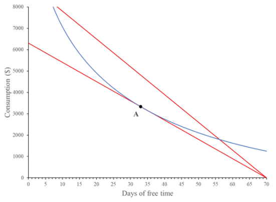
Income and substitution effects

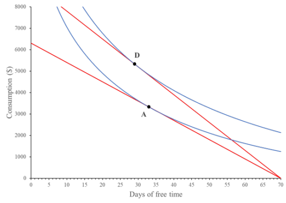
- Let's assume you are in the original situation; earning a daily wage of $90 and working 36 days (and consume $3 240)
- You hear about a new vacancy at the superstore that pays $130 per day and you get the job.
- Budget constraint now:
  - Consumption $c = 130(70 – d)$
- How will this affect your actions?

63

3.7

Income and substitution effects





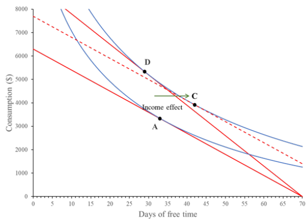
The budget constraint will intersect the consumption axis at $ 9 100 ($130 x 70)

New equilibrium is at **D** where MRT = MRS = w
We are going to break up the shift from A to d into an income effect and a substitution effect

64

3.7

Fig 3.13 Income and substitution effects

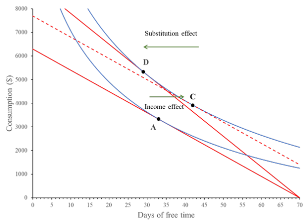


With an increase in wages, your wages increase and the budget constraint shifts parallel to the right
Income effect: Movement from **A** to **C**. Take more days off because your wage (income) is higher

65

3.7

Fig 3.13 Income and substitution effects

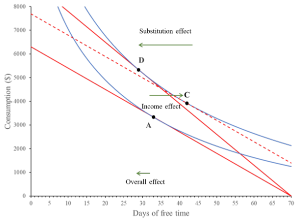


Due to the increase in wages, the opportunity cost of free time makes the budget constraint steeper
Substitution effect: The shift from **C** to **D**: You now choose to work more days and have less free days

66

3.7

Fig 3.13 Income and substitution effects



The net effect: Income effect + substitution effect
Movements are in opposite directions
Here substitution effect > income effect
Will thus work more days after the wage increase

67

Work hours: Income and substitution effects

Consumption and free time are **<u>normal goods</u>**: Demand increases as income rises.

<u>Income effect</u>	**<u>Substitution Effect</u>**
The higher income from every hour of work makes workers better off. It means that workers want to have more hours of free time, as well as more consumption.	The higher reward from working an extra hour has increased the opportunity cost of free time. This increases the incentive to work, and hence has the effect of reducing hours of free time.

Overall (price) effect = **Income effect** + **Substitution effect**
Which effect dominates depends on <u>individual preferences</u>

68

3.7

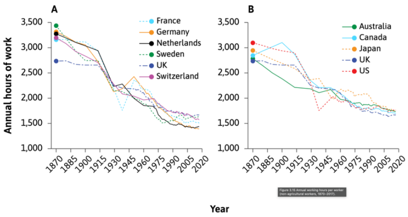
Is this model realistic?

- Worked with a model where the consumer chooses between consumption spending and free time
- The best point is where MRT = MRS
- Billiard player example
- Consumers do not make difficult mathematical calculations in decision making
- Economic theory helps us to predict behaviour
  - Ceteris paribus
  - Representative worker
  - Influence of culture and politics

69

3.9

Fig 3.15 Annual working hours per worker



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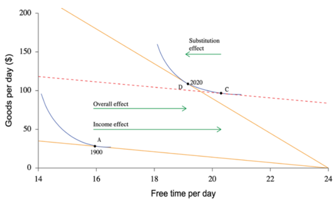
3.9

Fig 3.16 Income and substitution effects



Start at point A in 1990
Then wages increased and ended at point C in 2020
Income effect: The Shift from A to C with the increase in income has the effect of working fewer hours
Substitution effect: Rise in the opportunity cost of free time lets workers shift from C to D and work more hours
Net effect: Income effect > substitution effect. Workers worked fewer hours and consumed more

71

3.10

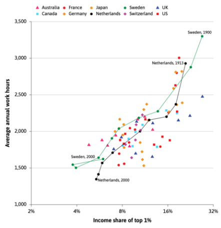
Working hours and inequality

- In previous section (Fig 3.15), we saw that over time people worked fewer hours
- Not only about the opportunity cost of free time. People may value free time more than the goods and services they consume
- Income became unequal and may be due to the rich that valued consumption more
- Conspicuous consumption: Signals to other members of society that they can afford the rich items
- Veblen effect: Other people work more hours to try to copy the standards of the rich

72

3.10

Fig 3.18 Working hours of the very rich



73

3.12

Differences between countries

- Workers with higher income tend to have more free time
- Big differences in annual hours of free time between countries of similar income
- Figure 3.25 represents the daily consumption and free time for the representative worker for specific countries
- The slope of the budget constraint corresponds to the wage

74

Is this a good model?

- Not realistic: People don't actually do **MRS/MRT** calculations. Most people cannot choose their working hours.
- BUT still a good approximation: Over time, people learn what combination of working hours and free time suits them best. Working hours can change due to culture and politics (indirect choice); people can choose which jobs to apply for.
- Helps us understand real-world phenomena: preferences and income/substitution effects can explain differences in working hours across countries and over time.

75

In the next unit

- Models of individual choice that include other important factors
- The role of **social interactions** in individual choice
- The effect of individual choice on **social outcomes**

76